

SF2324 Introduction to Applied Research Design and Quantitative Research Methods for Social Scientists, 15 credits

Course Plan

Instructors:

The person below marked with (*) is the course segment coordinator. Please remember to copy him in any queries or communication relevant to the course.

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Organization of the course:

The course is comprised of three blocks of teaching material and a final paper.

Block 1: Research Design

In this block, key components of research design are discussed. Focus is on causal inference, theory building and key designs (e.g experimental). The block culminates to a seminar and the first part of course work with the Research Design seminar (see below).

Block 2: Key concepts, Measurement, Central Tendency, Hypothesis testing part A

In this block, key concepts in statistics are discussed. The statistical software STATA is introduced and the first two labs of the course are run. In all computer labs, the students are encouraged to work in pairs.

Block 3: Regression Analysis, hypothesis testing part B

This block presents ordinary least squares (OLS), a workhorse for quantitative analysis. The logic and assumptions behind the regression analysis are presented. Regression diagnostics and some extensions are discussed. Two more labs will take place, and again, for the exercises students work in pairs.

The course is organized in lectures, seminars and labs. After a set of lectures either a seminar or a lab session will follow leading to an assignment. The course culminates into a final paper that will be examined in January 2025.

See the schedule below for the literature assigned to each lecture. A description of the coursework can also be found at the end of the document.

Learning Outcomes:

A student who has passed the course will be able to:

Knowledge and understanding

- Demonstrate comprehensive knowledge of fundamental problems within theory of science, relevant for various fields of science.
- Display comprehensive knowledge about the power and limitations of quantitative research methods.
- Critically assess the validity and reliability of secondary data and various data sources.
- Evaluate and independently design data collection methods such as surveys, field experiments or text data.
- Argue for the suitability of different research designs that fit particular research questions
- Understand and apply quantitative methods such as multivariate regression analysis.
- Confidently deploy and use statistical packages/software for analyzing data.

Competence and skills

- Specify, evaluate and test hypotheses and explanatory models applying quantitative methods.
- Present and clearly interpret results from quantitative analyses.
- Design and plan an independent research project using quantitative methods.
- Independently produce text in accordance with good academic practice, including proper citation technique and use of references.

Judgement and approach

- Independently evaluate conclusions from social science research based on statistical analysis
- Identify and assess ethical issues related to research on human behavior

Literature:

- Gerring John and Dino Christenson. 2017. Applied Social Science Methodology: an Introductory Guide. Cambridge University Press. 442 pages. **(GC)**
- Kellstedt, Paul and Guy Whitten. 2018. The Fundamentals of Political Science Research. Cambridge University Press. Third Edition 316 pages. **(KW)**
- King, Gary, Kehoane, Robert & Sidney Verba. 1994. Designing Social Inquiry. Princeton University Press, Princeton. **(KKV)**
- Lieberman, E. S., D. N. Posner, and L. L. Tsai. 2014. Does Information Lead to More Active Citizenship? Evidence from an Education Intervention in Rural Kenya. World Development 60:69-83. (available for download through the library)
- Brady, H. E. 2008. Causation and Explanation in Social Science. In J. Box- Steffensmeier, H. Brady, & D. Collier (Eds.). The Oxford Handbook of Political Methodology (217-70). New York: Oxford University Press (available for download through the library)
- Brambor, T, WR Clark, and M Golder. 2006. Understanding interaction models: Improving empirical analyses Political Analysis 14 (1), 63-82. (available for download through the library)

Schedule and Literature:

Note that the dates below can be subject to change due to unforeseen circumstances. **Consult the on line schedule which is always the correct and up to date schedule.** Once we know more about student numbers we will compile the groups. This usually happens 2-3 days after the course starts.

L=Lecture

S=Seminar

Lab= Computer Lab

Date		Theme	Literature
5/11	L1	Introduction to the course. Scientific research/Components of Research design	GC: ch.2 KW: ch.1 KKV: 1-34
6/11	L2	Causality, Theory Building, Causal Inference	KW: ch 3 GC: ch. 5, ch.6, <i>Recommended: Brady, 2008</i>
7/11	L3	Experimental Designs, Observational Studies	GC: ch 7, 118-124, 353-358 KW: ch 4 <i>Recommended: Lieberman et al., 2014</i>
10/11	L4	Key concepts and paper	KW: ch 5 (104-123), GC: ch 3 (31-46)
11/11	L/Lab	Introduction to STATA (split into groups)	Getting started in STATA Data management in STATA STATA's interface The STATA forum
12/11	L5	Measuring and summarizing data: levels, central tendency (mean median mode), variance and standard deviation (descriptive statistics)	KW: ch. 6 (125-142), GC: ch. 18 (282-291)
14/11	S	Evaluation of Research Design (based on Lectures 1-4)	<i>Information to be distributed at the start of the course</i>
17/11	L6	Probability, Normal distribution and z	KW: ch 7 (143-160), GC: ch 19 (292-301)
18/11	L7	Central limit theorem, characteristics of the sample mean, sampling distribution, SE of the mean, confidence interval, t	KW: ch 7 (143-160), GC: ch 20 (302-312)

20/11		Research Design Assignment due at 12.00	
21/11	Lab 1	Lab based on lectures L5-7	<i>Information available in Canvas at the start of the course</i>
24/11	L8	Central limit theorem, characteristics of the sample mean, sampling distribution, SE of the mean, confidence interval, t	KW: ch 7 (143-160), GC: ch 20 (302-312)
25/11		Lab 1 due at 14.00	
26/11	L9	Hypothesis testing: cross-tabs, t-test, correlation	KW: Ch 8 (161-185), CG: ch 21 (313-330) t-test: STATA annotated output correlation: STATA annotated output One-way ANOVA in STATA
28/11	Lab 2	Lab based on lectures L8-L9	<i>Information available in Canvas at the start of the course</i>
1/12	L10	Hypothesis testing 3. Regression bivariate 1	KW: 161-187, 188-195 <i>Recommended:</i> GC: 331-343 Bivariate regression: Stata annotated output
2/12		Lab 2 due at 14.00	
3/12	L11	Hypothesis testing 3. Regression bivariate 2	KW 195-207 <i>Recommended:</i> GC 331-343 Bivariate regression: Stata annotated output

04/12		Submission, final paper abstract 17.00	
5/12	Lab 3	Lab based on lectures L10-L11	<i>Information available in Canvas at the start of the course</i>
8/12	L12	Regression multivariate, Categorical predictors	KW ch. 10-11, GC 343-345, Multivariate regression: Stata annotated output Categorical Predictors: Stata output
9/12		Lab 3 due at 14.00	
9/12	S	Final Paper Abstracts Seminar (JS only)	
10/12	S	Final Paper Abstracts Seminar (NC, GX, LR)	
11/12	L13	Regression Assumptions - Diagnostics	KW 207-214, 258-272 Diagnostics: Stata output
11/12	L14	Regression Extensions: Interactions Terms	KW 256-258, Recommended: Brambor et al., 2003; Interactions: Stata example
12/12	Lab 4	Lab based mainly on lectures L12-L13 (Note: Likely that L13 might extend into the first half of L14)	<i>Information available in Canvas at the start of the course</i>

16/12		Lab 4 due at 14.00	
18/12		Individual Supervisions	Please note: Exact times will be arranged directly with the supervisor you will be assigned to via sign up
7/01		Drop in Supervisions	Please note: Available slots will open
13/01		Submission final paper 17.00	
16/1	S	Final Paper seminars	Discussants will be arranged after submission

Course work is divided into three parts:

1. Research Design Seminar and Research Design Paper

During the research design seminar students will present in groups of 2-4 (depending on the final numbers of the course) their evaluation of an academic paper that will be assigned to them at the start of the course. More instructions about the criteria that this evaluation should consist of will also become available on the introduction lecture.

After the seminar students are expected to hand in, individually, one written assignment with a similar scope based on a different set of academic papers. Students must choose 2 from the set of papers for the written assignment (these will also become available on line). Word limit for the whole exercise is 700-800 words. More instructions about the criteria that this evaluation should consist of will also become available on the introduction lecture.

Active participation in the Research Design seminar is required (participation is mandatory). All presentations are to be done on a presentation software like powerpoint. The Grade for the Research Design Assignment will be Pass (G) or Fail (U). **Students who miss the seminar will have to submit a make up exercise, 1500 words, providing an evaluation of three papers in order to receive a pass for that seminar.**

2. Lab Assignments:

There are four computer labs in the course segment and four different Lab assignments to be handed in. The students can work in pairs for the hand in to Labs 1 to 4. Students can use the time during the Lab to complete the exercise and/or complete it on their own time. **Please note that the lab assignments that will be found in canvas at the start of the course, are to about 95% the assignments as they will appear at each lab, but not necessarily exactly the same. For some or all of them we might need to do fine adjustments depending on the pace of the course, the needs of the students or other reasons.**

Students are required to hand in answers to all lab assignments (Labs 1 to 4). The Grade for the Lab Assignments will be Pass (G) or Fail (U). All lab and seminar assignments must receive a passing grade before credits for the course are given. Students that hand in lab reports on time, and that only need some extra work to receive a passing grade will have one chance to fix the remaining issues by handing in a revised version within two weeks after the first deadline. After this, reports will only be marked at the times of re-exams.

3. The course final paper:

In the course paper the students will, individually, design a study and conduct preliminary analyses using the tools provided throughout the course. The research design and analysis should build on available data.

Students will also have the opportunity to discuss their idea for the final paper with their instructors and colleagues in two separate occasions. After submitting the final paper abstract (i.e a short preliminary idea for their paper), there will be an abstract seminar where a group discussion of each idea will take place. In a later date students will have the opportunity to further discuss this idea with their instructors during individual supervision hours. Each student will be assigned one supervisor. About a week before the final paper deadline students will be given an opportunity via drop in sessions to discuss mostly issues related to data analysis. In this occasion students can choose available time slots but not necessarily the teacher. The papers will be presented at a final seminar (mandatory participation). Students will need to present their own paper but also discuss papers of their colleagues (these will be assigned to each student after submission). All presentations of papers are to be done on a presentation software like powerpoint. More information about the course paper and the final seminar will become available on line and during Lecture 4.

For reasons of fairness to everyone, only those students who submitted a final paper by the specified deadline will be able to take part in the final paper seminars. Students who will not participate at the final paper seminar will have to provide written discussions (of about 500 words each) of three final papers that will be assigned to them. Please note that late final papers will not be graded on a rolling basis but only after specified re-examination dates (please consult Canvas).

There is no specific word limit but we expect a paper that is no longer than 10 pages long (font size 12, 1.5 spacing) including main tables and graphs but excluding an appendix (where regression diagnostics and less important descriptive statistics can go) and bibliography list. The Grade for the final paper will be Pass (G), Pass with distinction (VG), or Fail (U). **Late final paper submissions, papers submitted after the original deadline, cannot receive a Pass with distinction (VG) but will only be graded as Pass (G) or Fail (U).**

The *overall course grade* will be on the Swedish grading scale of Pass (G), Pass with distinction (VG), or Fail (U). The course grade comes fully from the grade on course paper (provided a student has passed all seminar, and Lab assignments). Therefore, in order to receive a Pass with distinction (VG) for the course one has to receive a VG in the final paper.

Canvas

All materials for the course will be found on canvas. There will be separate modules for lecture materials and for each of the individual components of the course (research design seminar, Labs 1-4, abstracts and the final paper). There will also be folders for late assignments or make up

assignments (e.g when a mandatory seminar is missed).

Attempts to deceive during examinations

An attempt to deceive during examinations means that a student has attempted to give an impression of their performance in a written exam or some form of assessment that is not true. That is what is commonly referred to as cheating, or attempting to cheat. There are different kinds of attempts to deceive, such as plagiarism, prohibited aids and prohibited collaboration. Deception can also take other forms.

- **Plagiarism** means that a student has presented someone else's text or other material as if it were their own text or material.
- **Prohibited aid** means that a student has used aids, including AI, that were not permitted according to the instructions.
- **Prohibited collaboration** means that students have collaborated with each other in breach of the instructions.

Plagiarism: Copying material from the Internet, from books, journals, reports, or from a fellow student and then presenting it as one's own is strictly forbidden! Plagiarism is theft of someone else's work and will result in immediate fail of the assignment. Please note that the inclusion of a source in a bibliography list is not in itself sufficient attribution of another's work. In addition, unless it is a proper citation with a reference, any direct copying of text is forbidden. All submitted student work is screened with a program called *Ouriginal*, which compares a student's written work with published materials (including everything published on the internet) and all the work from prior students, to detect plagiarism. The material you have read and based your arguments and factual statements on should always be referenced using proper referencing technique. More information on references and how to avoid plagiarism can be found here:

<https://studentportal.gu.se/english/help-and-guidance/search-read-write/reference-management-and-plagiarism>

Prohibited aid: All your submitted texts should be your own original writing, and not written by someone else or generated by AI. Unless there are specific instructions for the use of AI for an assignment, it should not be used and presented as your own text. The use of any tools to cheat during on campus exams (e.g. DISA) is also strictly forbidden.

Prohibited collaboration: students can discuss their work, but individual submissions must be your own original writing, and hence not too similar the work of others (with the exception of group or joint assignments explicitly specified). Be extra careful to avoid using common notes from such collaborations.

Any kind of cheating on examinations is considered a serious offence. Proved or suspected cheating will be reported to the Disciplinary Committee of the University of Gothenburg, in accordance with the procedure of disciplinary matters at the department of political science.

The consequences may be severe for a student if found guilty. A student may be suspended from classes and campus for a determined time-period (from 3 weeks to half a year), which will result in a temporary suspension of all financial aid (including CSN) and may jeopardize a student's residence permit status. More information on disciplinary matters, including plagiarism and its

consequences, can be found here:

<https://studentportal.gu.se/english/study-environment-and-rules/disciplinary-matters/>